🔷 Classification Metrics

1️⃣ Confusion Matrix

Foundation of all classification metrics. Must understand this first.

	Predicted NO	Predicted YES
Actual NO	TN	FP
Actual YES	FN	TP

TP — True Positive: Model predicted YES and it was actually YES **TN — True Negative:** Model predicted NO and it was actually NO **FP — False Positive:** Model predicted YES but it was actually NO — Type 1 Error **FN — False Negative:** Model predicted NO but it was actually YES — Type 2 Error

Real example — Heart Disease:

TP — predicted disease, patient has disease ✅

TN — predicted no disease, patient is healthy ✅

FP — predicted disease, patient is actually healthy ❌

FN — predicted no disease, patient actually has disease ❌ DANGEROUS

FN is most dangerous in medical problems. Missing a sick patient.

2️⃣ Accuracy

Most basic metric. Percentage of correct predictions.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Example:

100 patients total

TP = 40, TN = 50, FP = 5, FN = 5

Accuracy = (40+50) / 100 = 90%

When to use: Balanced dataset — equal classes.

When NOT to use: Imbalanced dataset. If 95% are healthy — predicting everyone healthy gives 95% accuracy but is completely useless.

3️⃣ Precision

Of all patients predicted as positive — how many actually have disease?

Precision = TP / (TP + FP)

Example:

Model predicted 50 patients have disease

40 actually have disease, 10 are healthy

Precision = 40 / 50 = 80%

High precision means: When model says positive — it is usually right. **When to use:** When False Positive is costly. Example — spam detection. You don't want to mark important emails as spam.

4️⃣ Recall — Sensitivity

Of all patients who actually have disease — how many did model correctly identify?

Recall = TP / (TP + FN)

Example:

60 patients actually have disease

Model correctly identified 50 of them

Recall = 50 / 60 = 83%

High recall means: Model catches most actual positives. **When to use:** When False Negative is costly. Example — cancer detection. Missing a sick patient is dangerous.

For heart disease — Recall is more important than Precision.

5️⃣ F1 Score

Harmonic mean of Precision and Recall. Balances both.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Example:

Precision = 80%, Recall = 83%

F1 = 2 × (0.80 × 0.83) / (0.80 + 0.83) = 81.5%

When to use: Imbalanced dataset where both precision and recall matter.

F1 score is 0 when either precision or recall is 0. F1 score is 1 when both precision and recall are perfect.

🔷 Regression Metrics

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1 MAE — Mean Absolute Error

Average of absolute differences between predicted and actual values.

MAE = (1/n) × Σ|actual - predicted|

Example:

Actual: [100, 200, 300]

Predicted: [110, 190, 320]

Errors: [10, 10, 20]

MAE = (10+10+20) / 3 = 13.3

Properties:

- Same unit as target variable
- Not sensitive to outliers
- Easy to interpret
- Lower is better

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2 MSE — Mean Squared Error

Average of squared differences between predicted and actual values.

MSE = (1/n) × Σ(actual - predicted)^2

Example:

Errors: [10, 10, 20]

Squared: [100, 100, 400]

MSE = (100+100+400) / 3 = 200

Properties:

- Penalizes large errors heavily — squaring makes big errors bigger
- Not in same unit as target — squared
- Sensitive to outliers
- Lower is better

3️⃣ RMSE — Root Mean Squared Error

Square root of MSE. Brings error back to same unit as target.

RMSE = √MSE

Example:

MSE = 200

RMSE = √200 = 14.14

Properties:

- Same unit as target variable
- Penalizes large errors more than MAE
- Most commonly used regression metric
- Lower is better

RMSE vs MAE:

- If outliers are important — use RMSE
- If outliers should not dominate — use MAE

4️⃣ R2 Score — Coefficient of Determination

Measures how well model explains variance in data.

R2 = 1 - (SS_residual / SS_total)

SS_residual = $\Sigma(\text{actual - predicted})^2$

SS_total = $\Sigma(\text{actual - mean})^2$

Interpretation:

R2 = 1.0 → Perfect — model explains all variance

R2 = 0.8 → Model explains 80% of variance — good

R2 = 0.5 → Model explains 50% of variance — fair

R2 = 0.0 → Model is as good as predicting mean — useless

R2 < 0 → Model is worse than predicting mean

When to use: Always report R2 for regression problems. Easy to interpret.

For Unsupervised Learning Metrics

Unsupervised metrics are harder because there are no labels to compare against.

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1 Inertia — Within Cluster Sum of Squares

Sum of squared distances between each point and its cluster center.

Lower inertia = points are closer to their cluster center = tighter clusters

Used for: K-Means. Used in Elbow Method to find optimal K.

Limitation: Always decreases as K increases. Cannot use alone to pick best K.

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2 Silhouette Score

Measures how similar a point is to its own cluster compared to other clusters.

Score for each point = (b - a) / max(a, b)

a = average distance to points in same cluster

b = average distance to points in nearest other cluster

Best metric for clustering — use this to compare different K values.

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3 Davies-Bouldin Index

Measures average similarity between each cluster and its most similar cluster.

Lower Davies-Bouldin = better clustering

0 = perfect clustering

When to use: Comparing different clustering algorithms or different K values.

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4 Calinski-Harabasz Index — Variance Ratio

Ratio of between-cluster variance to within-cluster variance.

Higher score = better defined clusters

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5 Elbow Method — For finding K in K-Means

Not a metric itself but a technique using inertia.

Run K-Means for K = 1, 2, 3, 4, ... 10

Plot inertia for each K

Find the point where inertia stops decreasing sharply

That point looks like an elbow — that is the best K

🗣️ Interview Answers — Practice These

What metrics do you use for classification?

"For balanced datasets I use accuracy. For imbalanced datasets I use precision, recall and F1 score. For medical problems like heart disease I focus on recall because missing a sick patient is more dangerous than a false alarm. I also report AUC ROC score which gives threshold independent evaluation."

What is the difference between precision and recall?

"Precision is of all predicted positives how many are actually positive. Recall is of all actual positives how many did the model correctly identify. For spam detection precision matters more — don't want to lose important emails. For medical diagnosis recall matters more — don't want to miss sick patients."

What metrics do you use for regression?

"I use RMSE as primary metric because it is in same unit as target and penalizes large errors. I also report R2 score because it is easy to interpret — it tells what percentage of variance the model explains. If data has many outliers I use MAE instead of RMSE."

What is Silhouette Score?

"Silhouette score measures how well each point fits in its own cluster compared to other clusters. It ranges from -1 to 1. Score close to 1 means point is well matched to its cluster. Score close to 0 means point is on the boundary. Score close to -1 means point might be in wrong cluster. It is the best metric for evaluating clustering quality."